

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

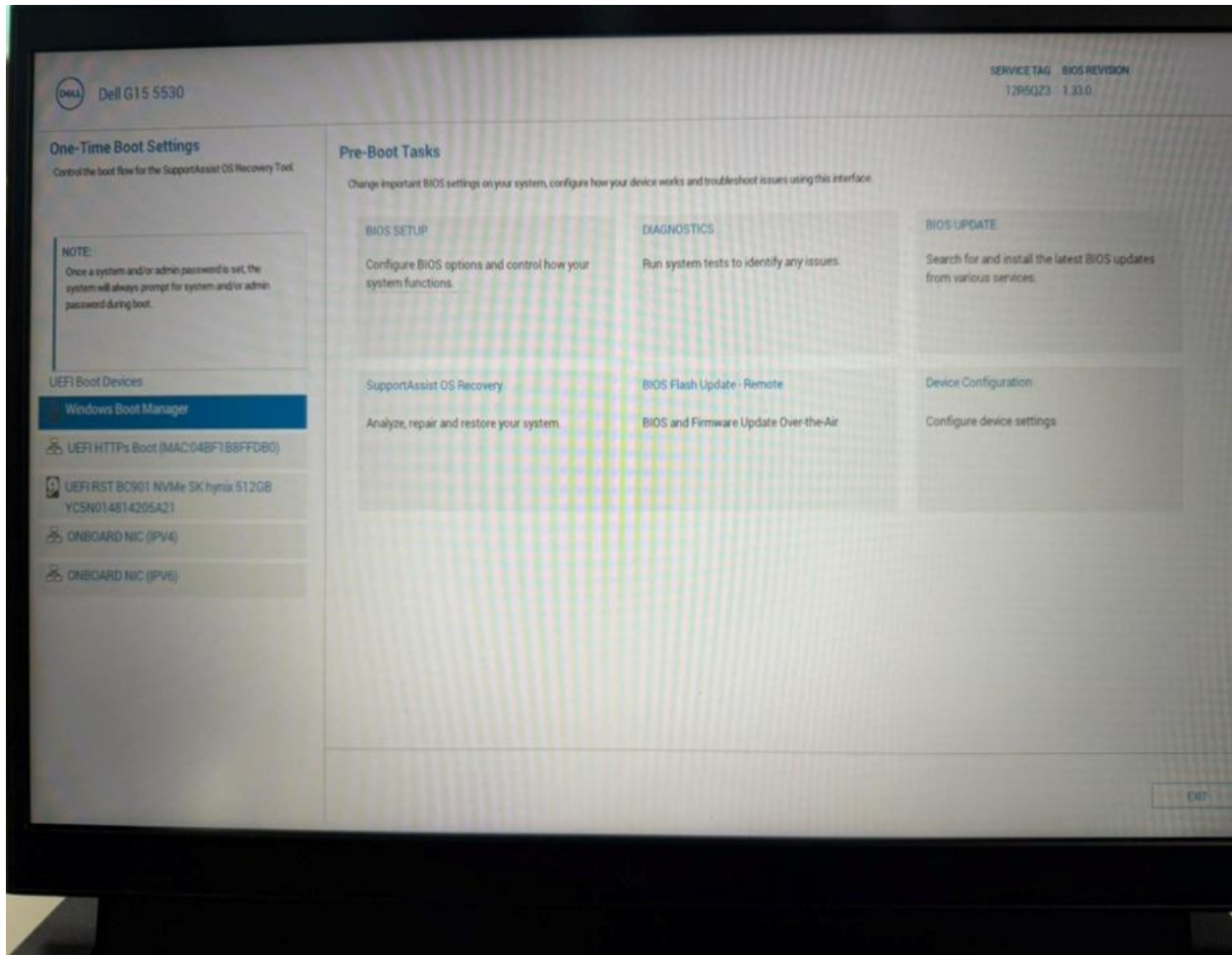
| BIOS/POST Error Message  | Usually Points To                                                   |
|--------------------------|---------------------------------------------------------------------|
| CMOS Checksum Error      | CMOS battery is weak or BIOS settings were reset.                   |
| Keyboard Not Detected    | Keyboard is not connected or is faulty.                             |
| No Boot Device Found     | Hard drive/SSD is not detected or no operating system is installed. |
| Memory Error (RAM Error) | RAM is loose, faulty, or not installed correctly.                   |
| CPU Fan Error            | CPU fan is not connected, not working, or spinning too slowly.      |

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

**Two troubleshooting steps:**

1. **Reseat the RAM** – Turn off the PC, unplug it, remove the RAM, and reinstall it securely.
2. **Test one RAM stick at a time** – If there are multiple RAM modules, try booting with one stick in each memory slot to identify a faulty RAM module or slot.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system. Hint: Think about the CMOS battery and BIOS settings reset

The message "CMOS checksum error – Defaults loaded" means that the BIOS settings stored in the CMOS memory have been lost or become corrupted. The computer loads the default BIOS settings because it cannot read the saved settings correctly. This is often caused by a weak or dead CMOS battery.

**How to Resolve the Issue:**

1. Restart the PC and enter the BIOS/UEFI Setup.
2. Check and correct the date, time, and BIOS settings if needed.
3. Save the settings and restart the computer.
4. If the error appears again, replace the CMOS battery (CR2032) with a new one.
5. After replacing the battery, set the correct BIOS settings again and save them.